MALDA

GAME DEVELOPMENT
CSCI - SENIOR CAPSTONE PROJECT

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Project Name

Malda is the main character's name and was also chosen for the game's title. Initially, the name was just a placeholder and came from how we described the gameplay. Our game was designed around a Mario + Zelda mashup concept, later incorporating elements from games like Hollow Knight and Terraria's visual aesthetics.

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Abstract

Malda is a 2D Puzzle-Platforming game that sees players navigate platforms, fight enemies, and overcome obstacles to reach their goal. Upon completing a level, players will be awarded a new item to help them continue their journey. Some levels require items from previous levels to be completed. After items have been unlocked, players will be able to access different areas of the level select hub, allowing access to more levels. Malda will feature a minimalistic pixel art style. Players will encounter multiple enemies as they progress through levels.

Malda players will need to be creative with their use of items, sometimes in combination with each other, in order to succeed. As players make it further in the game, they will find their technical platforming skills tested. Players will experience a couple of significant fights towards the end of the levels, creating a higher intensity environment for players to test their skills.

Tools & Technologies

Tool	Description	Usage
Unity	A full-fledged platform for creating 2D and 3D games and applications with the C# language.	The main platform to develop and test Malda. The free version will cover all bases for our purposes.
C#	An object-oriented programming language applicable to Unity development.	Development for Malda will be done with C#. The object-oriented nature and applicability with Unity integration make it a solid choice for development.
Github	A developer platform to consolidate projects that use Git version control.	GitHub allows for easier tracking of the project files, status, and versions. As our team utilizes Git, GitHub will be used to manage the project overall.
Git	An open source version control system to handle changes in project development.	Git will be the version control system tracking changes to Malda during development. Constant commits will be used to report changes in development, and pulls allow the team to stay on the same page with updated files.
Discord	A social platform for individual or group communication online.	Discord is the main communication platform for the development team. A server allows us to utilize instant messaging, as well as share and save important documents.
Google Gemini	An Artificial Intelligence tool that can be used for a variety of purposes.	Since our team has a Pro license for Google Gemini, it can be utilized for various aspects of development. Primarily, we will use it to generate pixel art and small sound files for Malda.
Gimp	Free and open source image editor.	Gimp is a free and easy-to-use image editor that will suffice for our team's sprite and background editing and creation needs.

Requirements List

1. Main Menu Screen

When players load up the game, a Main Menu screen will appear.

i. The game title will be in the center, top portion of the screen.
ii. The three main menu button options will be clickable.
iii. The main menu buttons will be in a collection centered on the screen and placed below the title of the game.

b. New World

The first button listed will be labeled as “New World” and will execute the following:

i. Starts a new game where the player will load into the level selection hub.
ii. The level selection hub will appear as a 2D platform, with levels 1-3 unlocked and playable.
iii. Levels 4 through 6 will show a visual cue letting the player know that they are not accessible at this time.

c. Load World

The second button, placed below the “New World” button, will be labeled as “Load World”. When clicking this button, it will execute the following:

i. Loads a single save file that will load the player into the level selection hub, where all previously saved events have occurred.
ii. This may include any items collected, secret areas unlocked, checkpoints reached, or levels that were locked initially but are now unlocked.
iii. See the level select hub for further details.

d. Game Settings

The third button will be labeled “Game Settings” and will load a new screen for the player once clicked.

i. The Game Settings menu will load onto the screen. And will have tabs on the side of the menu, starting from the top corner of the menu screen.
ii. **Tab 1** will be labeled as “Audio”. Header labeled “Audio settings”:

1. A slider labeled “Music Volume” that shows 0-100 depending on where the slider is placed.
    - This will be below the header text labeled “Audio settings”.
    - This will increase or decrease the volume of the background music only.
2. A slider labeled “Ambient Volume” that shows 0-100 depending on where the slider is placed.
    - This will be below the slider labeled “Music Volume”.
    - This will increase or decrease the natural sounds of the environment that the player is in.
3. A slider labeled “Overall Volume” that shows 0-100 depending on where the slider is placed
    - This will be below the slider labeled “Ambient Volume”.
    - This will increase or decrease all sounds that come from the game.

iii. **Tab 2** will be labeled “Graphics.”

1. Header text labeled as “Graphics Settings.”
2. A slider labeled “Brightness” that will dim or brighten the screen depending on where the slider is placed.
    - The slider “knob” will be placed in the center of the bar in its original position.
    - It will be below the header labeled “Graphics Settings”.
3. Header text labeled “Zoom.”
    - Two small buttons side by side will be placed under the header text “Zoom.”
    - The left button will be a “-” that zooms out to show more of the surrounding area.
    - The right button will be a “+” that zooms in, showing less of the surrounding area.
4. A drop-down window labeled “Resolution” will have a list of size options.
    - A radio button labeled “Windowed Mode” that, when checked, puts the game in a window vs. full screen.
        i. Windowed mode will automatically scale the game.
    - A radio button labeled “Full-Screen Mode” that, when checked, puts the game in full-screen mode.

            i. If Fullscreen is selected, a resolution dropdown will be populated, allowing players to choose between available monitor resolutions.

iv. **Tab 3** will be for keybinds, allowing players to adjust gameplay to their unique style.
    1. Key binds will have their original settings. As the document progresses, actions will be notated with their default keybinds, which will appear as a scrollable list here.
    2. Actions will be labeled on the left-hand side, while their respective keybound key will be labeled on the right-hand side of this menu screen.
        - Click the keybind label to change it to the preferred key.
        - Hit any key on the keyboard to assign it as the new key for that action. It will automatically save that key.

e. Quit

i. A button with the label “Quit” will have a small pop-up window in the center of the screen with a warning text reading “Warning: Any progress done may not have been saved at this point. Continue to Quit the game?”
    1. Button labeled “Yes” on the bottom left of the pop-up window
        - Will close and exit the game when clicked.
    2. Button labeled “No” on the bottom right of the pop-up window
        - Will return the player to their original screen.

2. Playable Mechanics

a. In Game Menu

i. The escape key will be the keybind for opening the in-game menu on all 6 levels.

ii. On the in-game menu screen, players will have 3 button options available to choose from:

iii. Button labeled “Return to Hub”:
    1. Leave the current level as is without completing it.
    2. Progress is not saved past any checkpoints reached.
        - Note: All changes to current gameplay, such as reaching new checkpoints on levels, will not be saved to the current and only available save file unless you

visit and save with the save NPC in the level select hub.

iv. Button labeled “Restart Level”:
    1. Selecting the “Restart Level” button restarts the entire level, returning players to the point where they first entered.

v. Button Labeled “Return to CheckPoint”
    1. When selected, Players will return to the most recently reached checkpoint in the current level.

b. Basic Movement

i. The player will be able to control the playable character's walking mechanics in-game.
    1. The player will use the A to move left in the 2D space. The playable character will only move while the key is being held down.
    2. The player will use the D key to move right in the 2D space. The playable character will only move while the key is being held down.

ii. The player will be able to control the playable character's running mechanics in-game.
    1. The player must hold down the shift key while also holding down the A key or the D key for faster movement; the playable character will only run while these combination keys are held down.

iii. The player will be able to use a jumping mechanic.
    1. The playable character will make a sound when jumping.
    2. The jumping mechanic requires the player to hit the space bar to jump.
    3. Double-pressing the space bar will trigger a double-jump mechanic.
    4. Pressing the space bar while holding down the combination keys for running will result in a higher jump, also giving the character further range before landing.

iv. The playable character will have a hitbox that emits sounds when colliding with other objects.

c. Items

i. **Wooden Sword**:
    1. The player will receive a basic wooden sword at the beginning of their gameplay in the level select hub.

2. Left-clicking the mouse initiates a swing attack.
3. Using the basic wooden sword is best for short-range attacks.
4. The sword will have a hit box that, if colliding with enemy hit boxes, will inflict damage upon the enemy.

ii. **Boomerang:**

1. A Boomerang will be received at some point during gameplay. See Level 1-6 for further details.
2. Mechanics include being thrown and returning to the player after reaching a certain distance.
3. Right-clicking the mouse will throw the boomerang-like weapon.
4. Using a ranged weapon is best for long-range attacks and reaching certain aspects of the levels that are initially unreachable by normal means.
5. A ranged weapon will have a hitbox, giving the ability to deal damage to enemies when colliding with their hitboxes.
    - Additional mechanics: Ranged-weapon hitbox will also provide the ability to activate some level objects for puzzle-solving aspects if the ranged-weapon hitbox collides with a world object hitbox.

iii. Climbing Boots:

1. Climbing boots will automatically be placed on once the item has been received.
2. Attaining boots will allow the player to climb up and down walls and jump off them at any time.
    - Boots will also allow jumping when the player is in contact with a wall.

iv. Orb of condensed slime:

1. This item will be obtained once defeating a slime enemy.
2. The orb is thrown onto a platform or onto the ground surface with the E key.
3. When in contact with the ground or platform surface, the area it lands on will be coated with “bouncy slime.”
    - Uses a hitbox to activate the bouncy slime coating mechanic.
        i. Players gain additional jump height when jumping or landing on top of the area coated with slime.

4. Pressing E again removes the slime, returning it to the orb in your inventory.
5. Leaving an area that still has slime coating placed down will automatically return the slime to the orb in your inventory.

v. **Gauntlets:**
1. The gauntlets are found in an early section of the Dwarven Realm.
2. They are used to push heavy objects to solve platforming level puzzles.
3. When obtained, they will automatically be equipped onto the character, instantly giving the player the “push heavy objects” buff.

vi. **Ice Beam Laser:**
1. Players will obtain an additional long-ranged weapon that shoots out an ice beam.
2. To shoot, play must press the F key.
   - Weapon’s discharge will have a larger hitbox.

d. Healing

i. Healing Items will be designed based on the level.
1. A health regeneration item can be found near enemies or hidden inside a breakable box.

ii. Other methods of healing include reaching a checkpoint.
1. When reaching a checkpoint, the player’s health will regenerate to full health.

iii. The player will have 3 Hitpoints to start with.
1. Health will appear as a horizontal bar serving as a counter at the top-right of the screen.
2. Completing a level will result in an increase in the player’s health by 1 hit point.

3. Enemy Mechanics

a. Grunts

i. Grunts are the universal, low-health enemies found throughout the game in each level.

ii. Grunts will have only 1 hit point.

iii. Grunts should be coded to move randomly.

iv. When a player comes within a small distance of a grunt enemy, the grunt will be “aggravated” and start moving toward the player to attack.
v. A grunt will attack with blunt weapons (stick or dull sword) at close range and deal 1 hit point of damage.

b. Turrets
    i. Enemies will be found in stationary positions throughout the levels; they will not move from their spots.
    ii. The enemy launches a projectile to either harm the player or damage certain areas nearby.
    iii. Turrets will have 3 hit points of health.

c. Slime
    i. Slimes will be dispersed throughout each level.
    ii. Slimes will not be triggered to attack the player when the player comes near them.
    iii. If a player collides with a slime, they will take 1 hit point of damage.
    iv. Slimes will seemingly wander aimlessly and bounce around as they move.

d. Dragon
    i. The dragon is level-specific; It is also the Big Bad Evil Guy on level 6.
    ii. The dragon will breathe fire.
        1. Fireballs will spew out of the dragon’s mouth like a short whip.
        2. The dragon will be stationary while in attack mode.

e. Ranged Grunt
    i. The ranger grunt will only have 1 hit point of health
    ii. The ranger grunt will shoot slowly with projectiles that deal 1 hit point of damage to the player if they collide with them.
    iii. Ranger grunts will be able to move around

f. Flying Imps
    i. Flies around above the players
    ii. Flying imps drop items onto the player.
        1. It will hold itself at its current position for 1 second before dropping the item.

4. Art

a. Player

i. 32x32 Sprite

ii. Idle Animation:
    1. 4 Frame sprite sheet
    2. Animation for the player when not moving
    3. This animation simulates breathing.

iii. Walking Animation:
    1. 4-Frame sprite sheet
        - One sprite sheet for the player when walking with a weapon in hand.
        - Another sprite sheet for the player walking without a weapon.

iv. Running Animation:
    1. 4 Frame sprite sheet
        - One sprite sheet for the player when running with a weapon in hand.
        - Another sprite sheet for the player running without a weapon.

v. Jumping Animation:
    1. 4 Frame sprite sheet
        - One sprite sheet for the player when jumping with a weapon in hand.
        - Another sprite sheet for the player jumping without a weapon.

vi. Player Hurt Animation:
    1. 1 Frame sprite sheet
    2. Player flashes red to indicate a hit

vii. Player Death Animation:
    1. 4 Frame sprite sheet
    2. The player falls over and dies.

viii. Throwing Animation:
    1. 4 Frame sprite sheet
    2. Specifically for throwing the orb of condensed slime.

b. Backgrounds

i. Level 1
    1. Level tile set: ground, edges, corners, fill tiles,
    2. Grassland/Farmland theme background

ii. Level 2

        1. Level tile set
        2. Forest theme background with heavy foliage
    iii. Level 3
        1. Level tile set
        2. Underground cavern theme background
    iv. Level 4
        1. Level tile set
        2. Mystical treetops theme background
    v. Level 5
        1. Level tile set
        2. Snowy, Icy mountainous theme background
    vi. Level 6
        1. Level tile set
        2. Top of the giant world tree with leaves and sky background
    vii. Hub
        1. Tile set
        2. Save NPC saving animation
        3. Vertical platforming with entrances leading to level zones
    viii. Main Menu Screen
        1. Malda title header
        2. Simple pixel art background
    ix. Game Menu Screen
        1. Simple pixel art Background
        2. Buttons and icons for settings and options

c. Platforms
    i. Various-sized platforms with distinct styles based on the zone
        1. Grass style platform
        2. Wood style platform
        3. Stone style platform
    ii. Temporary platform alteration with slime player object

d. Enemies
    i. Grunt
        1. 32x32 Sprite
        2. Walking animation
            - 4 Frames
        3. Melee attacking animation with a club
            - 4 Frames
        4. Hurt animation
            - 1 Frame
            - Flash red to indicate the enemy takes damage

ii. Turret
    1. 32x32 Sprite
    2. Shooting animation that fires a projectile
        - 4 Frames
    3. Hurt animation
        - 1 Frame
        - Flash red to indicate the enemy takes damage
iii. Slime
    1. 32x32 Sprite
    2. Walking animation
        - 4 Frames
    3. Bounce animation
        - 4 Frames
    4. Hurt animation
        - 1 Frame
        - Flash red to indicate the enemy takes damage
iv. Dragon
    1. Idle animation
        - 4 Frames
    2. Fireball ranged attack animation
        - 4 Frames
        - Shoots fireballs from the mouth towards the player
    3. Hurt animation
        - 1 Frame
        - Flash red to indicate the enemy takes damage
v. Ranged Grunt
    1. Walking animation
        - 4 Frames
    2. Ranged shooting attack animation
        - 4 Frames
        - Shoots an arrow projectile towards the player
    3. Hurt animation
        - 1 Frame
        - Flash red to indicate the enemy takes damage.
vi. Flying Imp
    1. Flying animation
        - 4 Frames
    2. Ranged attack animation
        - 4 Frames
        - Drops objects down on top of the player

            3. Hurt animation
                - 1 Frame
                - Flash red to indicate the enemy takes damage

e. Items

i. Wooden Sword
    1. 2-4 frames
    2. Swinging animation

ii. Orb of Condensed Slime
    1. Throwing animation
    2. Projectile animation
    3. Slimy Platform sprite

iii. Boomerang
    1. Throwing animation
    2. Animation for a boomerang in the air

iv. Ice Beam
    1. Shooting animation
    2. Projectile animation

v. Climbing Boots
    1. Wall jump animation

vi. Gauntlet
    1. Pushing animation

5. Sound

a. Background Music
    i. One standard 30-second-long music track
        1. Procedurally generated

b. Boss Music
    i. One standard 30-second-long music track
        1. Procedurally generated

c. Item Sounds
    i. Sword swing sound
    ii. Ice Beam shooting sound
    iii. Slime Ball
        1. Throwing sound
        2. Bouncing on slime sound
    iv. Boomerang throwing sound
    v. Wall jumping sound
    vi. Pushing sound

d. Damage Sounds

        i. Player gets hurt sound
        ii. Player hits the enemy sound
            1. Same regardless of how they hit the enemy
        iii. Death sound
            1. Somewhat similar for most enemies
            2. Player death 5-second track
    e. Movement Sounds
        i. One sound for steps
        ii. One sound for jumps
    f. Menu Sounds
        i. One basic menu sound for changing options

6. Level Select Hub

a. Theme: Tree of Life that simply exists in the cosmos.
b. There will be a sign by the hub entrance explaining the controls to the player.
c. Hub NPC used for saving the game:
    i. The NPC will initially be trapped near the hub entrance.
        1. The player will be able to pick up a sword next to the sign explaining controls.
        2. The player must swing their sword at the tree vines to set the save NPC free.
    ii. After saving the NPC, interacting with them will save the game
        1. Record levels that are completed on the local machine
d. Doors to different levels:
    i. Levels 1, 2, and 3 are immediately available

    ii. Levels 4 and 5 have obstacles requiring items from previous levels to access
        1. Level 4 requires a boomerang to access
        2. Level 5 requires climbing boots and gauntlets to access
    iii. Level 6 has obstacles requiring items from 4 and 5 to access
        1. Level 6 requires the orb of condensed slime and the ice beam to access

7. Level 1

a. Theme: Human realm
b. Tool received: Boomerang

c. Basic enemy: Slimes
d. Easy Obstacle
    i. Single slime encounter
e. Medium Obstacle
    i. Two slimes
    ii. Basic platforming
f. Hard Obstacle
    i. Discover something/someone valuable
        1. Slimes continue to attack until ten are defeated

8. Level 2

a. Theme: Elf Realm
b. Climbing Boots
c. Grunt
d. Easy Obstacle
    i. Jumping Puzzle
        1. Moving upward
e. Medium Obstacle
    i. Jumping Puzzles
        1. Two Routes
            - One facing enemies
            - The other is a harder platform/parkour path
    ii. Grunts waiting on platforms
        1. Ambush players
f. Hard Obstacle
    i. Boss Fight
    ii. The enemy keeps changing sides of the arena
    iii. Players have to traverse platforms to continue fighting

9. Level 3

a. Dwarf realm
b. Strength gauntlets are used for moving heavy objects
    i. Players receive at the beginning of the level
c. Turrets
    i. Doesn’t move
    ii. Shoots in place
d. Easy Obstacle
    i. Players must evade the line of sight of a turret

e. Medium Obstacle
    i. Two Turrets
    ii. Moveable Blocks
    iii. Players must move blocks to avoid the sight of turrets
f. Hard Obstacle
    i. Obstacle in the player's way
        1. The turret can break the obstacle
    ii. Blocks available to move
        1. Start out blocking turrets
        2. Players must move the blocks and the trick turret to break the obstacle

10. Level 4

a. Theme: Fairy realm
b. Orb of condensed slime
c. Slimes
    i. Leave bouncy slime after death
d. Flying imps
    i. The player can kill after using bouncy slime
e. Easy Obstacle
    i. Evade flying imps through a platforming section
f. Medium Obstacle
    i. Players must climb a ladder
        1. Imps would stop the player if not killed
    ii. Slimes continuously spawn
    iii. The player must kill the flying imp in order to climb the ladder and continue
g. Hard Obstacle
    i. More technical platforming section
    ii. Multiple flying imps are attacking the player

11. Level 5

a. Dragon realm
b. Dragon Boss
c. Ice beam
d. Easy
    i. Dragon destroys platforms
    ii. The player must take an alternate route

e. Medium
    i. Dragon attacks on a timer
    ii. Player from platform to platform across the room to get to the dragon
f. Hard
    i. Player acquires ice beam
    ii. Player fights dragon
        1. The player must use the ice beam when the dragon breathes fire

12. Level 6

a. Decrepted/tainted realm
b. Final boss
    i. 3 hits to kill
    ii. Players have to climb up platforms to damage
        1. Fall back down after a hit
    iii. Players must use all items to traverse this level
    iv. Two hands as enemies to kill on the way to the boss's hit box